

# Real-World Problems and Significant Digits

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## MA.8.NSO.1.6

*I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.*

*Yesterday you multiplied and divided in scientific notation.*

**Today the numbers are measurements — and a measurement only knows so much.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1.  $(4 \times 10^5)(3 \times 10^{-2})$
2.  $(6.3 \times 10^7) - (4.1 \times 10^6)$
3. Write 0.00902 in scientific notation.
4. Round 4.738 to the nearest hundredth.

# Warm-Up

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*Answers.*

1.  $(4 \times 10^5)(3 \times 10^{-2})$        **$1.2 \times 10^4$**
2.  $(6.3 \times 10^7) - (4.1 \times 10^6)$        **$5.89 \times 10^7$**
3. Write 0.00902 in scientific notation.       **$9.02 \times 10^{-3}$**
4. Round 4.738 to the nearest hundredth.      **4.74**

# Notes

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*Copy the table.*

## I. Which digits are significant

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| Measurement | Digits | Why                                        |
|-------------|--------|--------------------------------------------|
| 4,207       | 4      | the middle zero counts                     |
| 0.0058      | 2      | leading zeros never count                  |
| 3.200       | 4      | trailing zeros count after a decimal point |
| 52,000      | 2      | no decimal point, so they do not           |

# Notes

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## II. Multiplying or dividing: count the digits

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$$(2.5 \times 10^2)(4.13 \times 10^2) = 1.0325 \times 10^5$$

2 significant digits and 3, so the answer keeps 2:  $1.0 \times 10^5$

**Multiply or divide: the answer takes the smaller of the two significant-digit counts.**

# Notes

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*Copy all three lines.*

## III. Adding or subtracting: a different rule

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$$1,234 + 5.6 = 1,239.6$$

1,234 stops at the ones place; 5.6 stops at the tenths

so the sum stops at the ones place:  $1,240 = 1.240 \times 10^3$

**Add or subtract: the answer stops at the leftmost of the two last-digit places.**

# Example 1

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*A product of two measurements.*

A solar array measures  $1.25 \times 10^2$  m by  $8.6 \times 10^1$  m.

What is its area?

**Give the answer to the correct number of significant digits.**

# Example 1

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*Worked.*

$$(1.25 \times 10^2)(8.6 \times 10^1) = 10,750 \quad \text{multiply in full}$$

$$1.25 \rightarrow 3 \text{ digits}, \quad 8.6 \rightarrow 2 \text{ digits} \quad \text{the answer keeps 2}$$

$$10,750 \rightarrow 11,000 = 1.1 \times 10^4 \quad \text{round once, at the end}$$

**Answer:  $1.1 \times 10^4$  square metres**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

A field measures  $4.6 \times 10^2$  m by  $2.35 \times 10^2$  m.

What is its area?

# Your Turn

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*Answer.*

A field measures  $4.6 \times 10^2$  m by  $2.35 \times 10^2$  m.

What is its area?

108,100, kept to two significant digits

**Answer:  $1.1 \times 10^5$  square metres**

## Example 2

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*A difference of two measurements.*

A container holds  $3.09 \times 10^{-3}$  cubic feet of liquid.

$3.472 \times 10^{-5}$  cubic feet evaporates.

**How much is left? Give the answer with the correct precision.**

## Example 2

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*Worked.*

$$0.00309 - 0.00003472 = 0.00305528 \quad \text{subtract in full}$$

$$0.00309 \rightarrow \text{stops at } 10^{-5} \quad \text{the leftmost last-digit place}$$

$$0.00305528 \rightarrow 0.00306 = 3.06 \times 10^{-3} \quad \text{round there}$$

**Answer:  $3.06 \times 10^{-3}$  cubic feet**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

A road is  $3.85 \times 10^3$  m long.

A section  $2.74 \times 10^2$  m long is repaved.

How much of the road is not repaved?

# Your Turn

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*Answer.*

A road is  $3.85 \times 10^3$  m long.

A section  $2.74 \times 10^2$  m long is repaved.

How much of the road is not repaved?

3,576, stopped at the tens place

**Answer:  $3.58 \times 10^3$  m**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

How many significant digits are in 0.00470 ?

**Answer it.**

*Leading zeros, then trailing zeros.*

# Whiteboards • Question 1 of 9

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*Answer.*

How many significant digits are in 0.00470 ?

4, 7 and the last 0

**Answer: 3**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

A pond's surface measures  $1.4 \times 10^2$  m by  $8.5 \times 10^1$  m.

The pond is 3 metres deep.

What is the surface area, to the right number of significant digits?

**Answer it.**

*Two digits each.*

# Whiteboards • Question 2 of 9

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*Answer.*

A pond's surface measures  $1.4 \times 10^2$  m by  $8.5 \times 10^1$  m.

The pond is 3 metres deep.

What is the surface area, to the right number of significant digits?

11,900, kept to two digits

**Answer:  $1.2 \times 10^4$  square metres**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is  $7.4126 \times 10^5$ .

How should the answer be written?

**Answer it.**

*Which count wins?*

# Whiteboards • Question 3 of 9

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*Answer.*

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is  $7.4126 \times 10^5$ .

How should the answer be written?

the smaller count

**Answer:  $7.41 \times 10^5$**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$(2.5 \times 10^3) + (4.71 \times 10^2)$  metres, both measured.

Write the sum with the correct precision.

**Answer it.**

*Where does each one's last digit sit?*

# Whiteboards • Question 4 of 9

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*Answer.*

$(2.5 \times 10^3) + (4.71 \times 10^2)$  metres, both measured.

Write the sum with the correct precision.

2,500 stops at the hundreds

**Answer:  $3.0 \times 10^3$  m**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

How many significant digits are in 0.03080 ?

**A.** 4

**B.** 3

**C.** 5

**D.** 2

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

How many significant digits are in 0.03080 ?

**A. 4**

**B. 3**

**C. 5**

**D. 2**

**Answer: A — 4**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

A rectangle measures  $6.0 \times 10^2$  cm by  $4.35 \times 10^2$  cm.

Which is the area, written correctly?

**A.**  $2.6 \times 10^5 \text{ cm}^2$

**B.**  $2.61 \times 10^5 \text{ cm}^2$

**C.**  $26.1 \times 10^4 \text{ cm}^2$

**D.**  $2.6 \times 10^4 \text{ cm}^2$

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

A rectangle measures  $6.0 \times 10^2$  cm by  $4.35 \times 10^2$  cm.

Which is the area, written correctly?

**A.  $2.6 \times 10^5 \text{ cm}^2$**

**B.  $2.61 \times 10^5 \text{ cm}^2$**

**C.  $26.1 \times 10^4 \text{ cm}^2$**

**D.  $2.6 \times 10^4 \text{ cm}^2$**

**Answer: A —  $2.6 \times 10^5 \text{ cm}^2$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

A car travels  $4.20 \times 10^2$  km in 3.0 hours.

What is its average speed, to the right number of significant digits?

**Answer it.**

*Count both measurements.*

# Whiteboards · Question 7 of 9

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*Answer.*

A car travels  $4.20 \times 10^2$  km in 3.0 hours.

What is its average speed, to the right number of significant digits?

140 exactly, kept to two digits

**Answer:  $1.4 \times 10^2$  km/h**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

A measured length times a measured width gives an area of  $3.6 \times 10^3 \text{ m}^2$ . The length measured  $9.0 \times 10^1 \text{ m}$ .

What did the width measure, to two significant digits?

**Answer it.**

*Divide, then count.*

# Whiteboards • Question 8 of 9

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*Answer.*

A measured length times a measured width gives an area of  $3.6 \times 10^3 \text{ m}^2$ . The length measured  $9.0 \times 10^1 \text{ m}$ .

What did the width measure, to two significant digits?

$$3,600 \div 90$$

**Answer:  $4.0 \times 10^1 \text{ m}$**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,  
so the sum  $1,234 + 5.6$  must be written with 2:  $1.2 \times 10^3$ .

Explain why he is wrong, and give the correct sum.

**Write your answer in sentences.**

*Which rule is for which operation?*

# Whiteboards · Question 9 of 9

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*Answer.*

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,  
so the sum  $1,234 + 5.6$  must be written with 2:  $1.2 \times 10^3$ .

Explain why he is wrong, and give the correct sum.

products count digits; sums count places

**Answer:  $1,240 = 1.240 \times 10^3$**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S